Lecture 12

Homogeneous spaces and compact induction

Objectives. *Let K be a compact Lie group and H a closed subgroup. Which irreducible representations of K occur in functions on K/H, and how often? We show that each irreducible representation V of K occurs $\dim V^H$ times. Induction from H extends this count to vector-valued functions.*

By the end of this lecture you will be able to:

- Construct the invariant integral on K/H and identify $L^2(K/H)$ with the right-H-invariant part of $L^2(K)$.
- Prove that each irreducible V occurs in $L^2(K/H)$ exactly $\dim V^H$ times.
- Define compact induction, describe an element of the induced representation in the tangent-field model on S^2, and prove Frobenius reciprocity with finite multiplicities.
- Decompose $L^2(S^2)$ under $\mathrm{SO}(3)$ and relate the summands to spherical harmonics.

Throughout, K is a compact Lie group, not necessarily connected, and $H \subseteq K$ is a closed subgroup. By Lecture 3, K/H is a compact manifold, and the projection $p\colon K \to K/H$, $p(x) = xH$, is continuous and open. The group K acts on K/H by $g \cdot xH = gxH$.

The Haar measures dx on K and dh on H have total mass 1. They are left, right and inversion invariant (Lecture 5). The space $L^2(K)$ is the completion of $C(K)$ for $\langle f_1, f_2\rangle = \int_K \overline{f_1}\, f_2\, dx$, and $C(K) \to L^2(K)$ is injective (Lecture 10).

We use the left and right translation representations

$$(L_g f)(x) = f(g^{-1}x), \qquad (R_g f)(x) = f(xg).$$

These commuting representations are unitary and strongly continuous (Lecture 10).

12.1 Functions and integration on K/H

By the definition of the quotient topology, a function φ on K/H is continuous exactly when $\varphi \circ p$ is continuous. A right-H-invariant function on K is constant on cosets. Hence the pullback p^* is a linear bijection

$$p^*\colon C(K/H) \to C(K)^H := \{f \in C(K) : f(xh) = f(x) \text{ for all } x \in K,\ h \in H\}, \qquad p^*\varphi = \varphi \circ p. \tag{12.1}$$

Definition 12.1. The *invariant integral* on K/H is the linear functional

$$\int_{K/H} \varphi(\xi)\, d\xi := \int_K \varphi(p(x))\, dx, \qquad \varphi \in C(K/H).$$

Proposition 12.2. *(a) The integral is positive, $\int_{K/H} 1\, d\xi = 1$, and $\int_{K/H} \varphi(g^{-1}\xi)\, d\xi = \int_{K/H} \varphi(\xi)\, d\xi$ for $g \in K$.*

(b) For $f \in C(K)$, the H-average $f^H(x) = \int_H f(xh)\, dh$ lies in $C(K)^H$, and

$$\int_K f(x)\, dx = \int_K f^H(x)\, dx = \int_{K/H} \Big(\int_H f(xh)\, dh \Big)\, d(xH).$$

Proof. (a) Positivity and normalisation hold for dx. Invariance follows from $\varphi(g^{-1}p(x)) = \varphi(p(g^{-1}x))$ and left invariance of dx.

(b) The map $(x,h) \mapsto f(xh)$ is uniformly continuous on the compact space $K \times H$. So f^H is continuous. For $k \in H$, left invariance of dh gives $f^H(xk) = \int_H f(xkh)\, dh = f^H(x)$. Fubini for continuous functions on $K \times H$ and right invariance of dx give

$$\int_K f^H(x)\, dx = \int_H \int_K f(xh)\, dx\, dh = \int_H \int_K f(x)\, dx\, dh = \int_K f(x)\, dx.$$

By the pullback bijection (12.1), the last expression in (b) is $\int_K f^H\, dx$ written on K/H. ∎

A positive normalised invariant integral I on K/H is unique. Indeed, the map $g \mapsto \varphi(g^{-1}\,\cdot\,)$ is continuous in the supremum norm. So averaging the identity $I(\varphi) = I(\varphi(g^{-1}\,\cdot\,))$ over K gives $I(\varphi) = I\big(\int_K \varphi(g^{-1}\,\cdot\,)\, dg\big)$. The function inside is constant, since K acts transitively on K/H, and its value is $\int_{K/H} \varphi\, d\xi$. In particular, for a maximal torus $H = T$ of a compact connected group K, this integral is the measure $d(gT)$ of Lecture 9.

Definition 12.3. The space $L^2(K/H)$ is the completion of $C(K/H)$ for $\langle \varphi, \psi \rangle = \int_{K/H} \overline{\varphi}\, \psi\, d\xi$. Also

$$L^2(K)^H = \{ f \in L^2(K) : R_h f = f \text{ for all } h \in H \}.$$

The form is definite since $\int_{K/H} |\varphi|^2\, d\xi = \|\varphi \circ p\|^2$ and $C(K) \to L^2(K)$ is injective. The same identity shows that p^* is isometric. So p^* extends to an isometry $L^2(K/H) \to L^2(K)$. The subspace $L^2(K)^H$ is closed because each R_h is bounded.

The operator of the next proposition replaces a function by its average over each coset. Averaging does not increase an L^2 norm, and it does not change a function that is constant on cosets. In the sense of the vector-integral lemma of Lecture 10, this operator is the integral $\int_H R_h\, dh$. So it is the projector onto invariants of Lecture 5, on an infinite-dimensional space.

Proposition 12.4. *There is a unique bounded operator A_H on $L^2(K)$ with $A_H f = f^H$ for $f \in C(K)$. It satisfies*

$$\langle f_1, A_H f_2 \rangle = \int_H \langle f_1, R_h f_2 \rangle\, dh \qquad (f_1, f_2 \in L^2(K)). \tag{12.2}$$

It is the orthogonal projection onto $L^2(K)^H$. Consequently p^* maps $L^2(K/H)$ isometrically onto $L^2(K)^H$.

Proof. Let $f \in C(K)$. Cauchy–Schwarz on H gives $|f^H(x)|^2 \le (|f|^2)^H(x)$. Integrate and apply the averaging identity of Proposition 12.2(b) to $|f|^2$:

$$\|f^H\|^2 \le \int_K (|f|^2)^H \, dx = \int_K |f|^2 \, dx = \|f\|^2.$$

Since $C(K)$ is dense, A_H exists uniquely, with $\|A_H\| \le 1$.

For continuous f_1, f_2, Fubini shows that both sides of (12.2) equal $\int_K \int_H \overline{f_1(x)} f_2(xh) \, dh \, dx$. Both sides are continuous in (f_1, f_2). On the right, the integrand is continuous in h by strong continuity and bounded by $\|f_1\| \, \|f_2\|$. So (12.2) holds on $L^2(K)$.

Now $R_h^* = R_{h^{-1}}$ and dh is inversion invariant. So the weak formula (12.2) gives $\langle f_1, A_H f_2 \rangle = \langle A_H f_1, f_2 \rangle$, and A_H is self-adjoint.

Next, A_H fixes $L^2(K)^H$. If $f \in L^2(K)^H$, then the weak formula gives $\langle f_1, A_H f \rangle = \langle f_1, f \rangle$ for all f_1, so $A_H f = f$.

Finally, the range of A_H lies in $L^2(K)^H$. Let $k \in H$. For continuous f, $A_H f = f^H$ lies in $C(K)^H$ by the description of the H-average in Proposition 12.2(b), so $R_k A_H f = A_H f$. Since $C(K)$ is dense and both sides are bounded in f, this identity holds for all f.

Hence A_H is a self-adjoint idempotent with range $L^2(K)^H$.

It remains to identify the image of p^*. Since p^* is an isometry, its image is the closure of $p^*(C(K/H)) = C(K)^H$. This closure lies in the closed subspace $L^2(K)^H$. Conversely, let $f \in L^2(K)^H$ and choose $f_n \in C(K)$ with $f_n \to f$. Then $f_n^H \in C(K)^H$ and $f_n^H = A_H f_n \to A_H f = f$. So the image is $L^2(K)^H$. ∎

Each L_g commutes with every R_h, so L_g preserves $L^2(K)^H$. Transported by p^*, the operator L_g acts on $C(K/H)$ by $\varphi \mapsto \varphi(g^{-1} \cdot)$. The resulting unitary representation of K on $L^2(K/H)$ is the *left regular representation*.

12.2 Multiplicities in $L^2(K/H)$

Let (Π, V) be a finite-dimensional unitary representation and $u, v \in V$. The *matrix coefficient* is $c_{u,v}(x) = \langle u, \Pi(x) v \rangle$. Unitarity gives

$$L_g c_{u,v} = c_{\Pi(g)u,\, v}, \qquad R_g c_{u,v} = c_{u,\, \Pi(g)v}. \tag{12.3}$$

For the first identity, $c_{u,v}(g^{-1}x) = \langle u, \Pi(g)^{-1} \Pi(x) v \rangle = \langle \Pi(g) u, \Pi(x) v \rangle$.

The map $u \mapsto c_{u,v}$ is antilinear. The *conjugate representation* $\overline{V}$ is the set V with scalar multiplication $t \cdot u = \bar{t} u$, the same operators $\Pi(g)$, and inner product $\langle u, u' \rangle_{\overline{V}} = \langle u', u \rangle$. It has the same invariant subspaces as V. So $\overline{V}$ is irreducible when V is, and $(\overline{V})^H = V^H$ as sets.

For each $\sigma \in \mathrm{Irr}(K)$ fix a unitary representative (Π_σ, V_σ) of dimension d_σ. Write $\bar{\sigma}$ for the class of $\overline{V_\sigma}$, and let $E_\sigma = \mathrm{span}\{c_{u,v} : u, v \in V_\sigma\} \subseteq C(K)$. We use two earlier results:

(PW1) Schur orthogonality (Lecture 5): $\langle c_{u,v}, c_{u',v'} \rangle = d_\sigma^{-1} \langle u', u \rangle \langle v, v' \rangle$ for $u, v, u', v' \in V_\sigma$.

(PW2) Peter–Weyl (Lecture 10): the E_σ are mutually orthogonal and $L^2(K) = \widehat{\bigoplus}_\sigma E_\sigma$.

By Schur orthogonality (PW1), the linear map $\overline{V_\sigma} \otimes V_\sigma \to E_\sigma$, $u \otimes v \mapsto c_{u,v}$, is $d_\sigma^{-1/2}$ times an isometry. So it is bijective. By the coefficient identities (12.3), it carries $\bar{\Pi}_\sigma \otimes 1$ to L and $1 \otimes \Pi_\sigma$ to R. Let Q_σ be the orthogonal projection onto E_σ. A unitary operator that preserves a finite-dimensional subspace also preserves its orthogonal complement. Hence Q_σ commutes with all L_g and R_g. This Q_σ is the projector onto the matrix coefficients of Π_σ, called P_{Π_σ} in Lecture 11.

For $\sigma \in \mathrm{Irr}(K)$ let $E_\sigma^H = \mathrm{span}\{c_{u,v} : u \in V_\sigma,\ v \in V_\sigma^H\}$, a subspace of E_σ.

We read the count from the matrix coefficient $c_{u,v}$. Right translation by H acts only on v, so $c_{u,v}$ is right-H-invariant when $v \in V_\sigma^H$. Left translation acts only on u, and u enters antilinearly. So the right-H-invariant part of E_σ is $\dim V_\sigma^H$ copies of $\overline{V_\sigma}$. Block σ therefore contributes to the multiplicity of $\bar{\sigma}$. Since $\dim V_\sigma^H = \dim V_{\bar{\sigma}}^H$, the count for σ is still $\dim V_\sigma^H$. For SO(3) every irreducible is self-conjugate, and no shift occurs.

Figure 12.1 draws this count in the arrays of functions of Lecture 10. In a block, the entry c_{e_i,e_j} has its second index on the side of R. So the condition $f(xh) = f(x)$ keeps the columns whose vector e_j lies in V_σ^H.

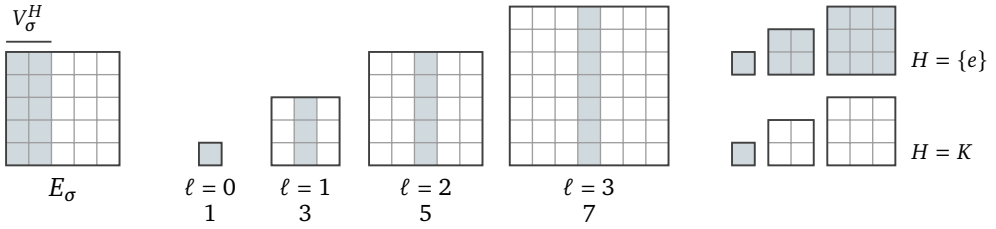


Figure 12.1: Blocks of functions in $L^2(K)$, as in Lecture 10, with the columns that remain on K/H shaded. From left to right: a general H, in a basis of V_σ whose first vectors span V_σ^H; the sphere SO(3)/SO(2), with blocks of side $2\ell + 1$ and one column each, of $2\ell + 1$ functions, for the weight-zero vector of § 12.4; and the extreme cases $H = \{e\}$ and $H = K$. Left translation mixes the entries of a shaded column.

Proposition 12.5. *Let $P_\sigma = \int_H \Pi_\sigma(h)\, dh$ be the orthogonal projection of V_σ onto V_σ^H (Lecture 5). Then $A_H c_{u,v} = c_{u,P_\sigma v}$. Moreover*

$$L^2(K/H) \cong L^2(K)^H = \overline{\bigoplus_{\sigma \in \mathrm{Irr}(K)}} E_\sigma^H, \qquad E_\sigma^H \cong \overline{V_\sigma} \otimes V_\sigma^H,$$

where K acts on $\overline{V_\sigma} \otimes V_\sigma^H$ by $\bar{\Pi}_\sigma \otimes 1$.

Proof. The finite-dimensional integral commutes with the functional $w \mapsto \langle u, \Pi_\sigma(x) w \rangle$. So

$$(A_H c_{u,v})(x) = \int_H \langle u, \Pi_\sigma(x)\Pi_\sigma(h) v \rangle\, dh = \left\langle u, \Pi_\sigma(x) \int_H \Pi_\sigma(h) v\, dh \right\rangle = c_{u,P_\sigma v}(x).$$

Thus $A_H(E_\sigma) = E_\sigma^H$, and A_H fixes E_σ^H. Hence $E_\sigma^H \subseteq L^2(K)^H$. Conversely, let $f \in L^2(K)^H$. By Peter–Weyl (PW2), $f = \sum_\sigma Q_\sigma f$ in L^2. Since A_H is bounded,

$$f = A_H f = \sum_\sigma A_H Q_\sigma f, \qquad A_H Q_\sigma f \in E_\sigma^H.$$

Since E_σ^H lies in E_σ, the spaces E_σ^H are mutually orthogonal by Peter–Weyl (PW2). This proves the decomposition. The isomorphism with $\overline{V_\sigma} \otimes V_\sigma^H$ is the restriction of the bijection above. ∎

Induction in § 12.3 uses vector-valued functions and evaluates them at the identity e. An L^2 class has no value at a point. The next lemma shows that equivariant images of V_σ consist of continuous functions, so these evaluations make sense.

Let W be a finite-dimensional Hilbert space with orthonormal basis $w_1, \dots, w_n$. Then $L^2(K) \otimes W$ is the Hilbert space of functions $\sum_j f_j w_j$ with $f_j \in L^2(K)$. It contains $C(K, W) = C(K) \otimes W$. Let K act by $L \otimes 1$.

Lemma 12.6. *Let $\sigma \in \operatorname{Irr}(K)$. Let $T : V_\sigma \to L^2(K) \otimes W$ be linear with $T\Pi_\sigma(g) = (L_g \otimes 1)T$ for all $g \in K$. Then $T(V_\sigma) \subseteq E_{\bar\sigma} \otimes W$. In particular, every vector in $T(V_\sigma)$ is a continuous function.*

Proof. Fix $\eta \in \operatorname{Irr}(K)$. The operator $Q_\eta \otimes 1$ commutes with $L_g \otimes 1$. So $(Q_\eta \otimes 1)T$ is a K-map from V_σ into

$$E_\eta \otimes W \cong \overline{V_\eta} \otimes (V_\eta \otimes W),$$

where K acts only on the first factor. This space is a direct sum of $d_\eta \dim W$ copies of $\overline{V_\eta}$. By Schur's lemma, $(Q_\eta \otimes 1)T = 0$ unless $V_\sigma \cong \overline{V_\eta}$, that is, unless $\eta = \bar\sigma$. Apply Peter–Weyl (PW2) in each coordinate. For $v \in V_\sigma$,

$$Tv = \sum_\eta (Q_\eta \otimes 1)Tv = (Q_{\bar\sigma} \otimes 1)Tv \in E_{\bar\sigma} \otimes W.$$

∎

Thus the values $(Tv)(x)$ are defined for all $v \in V_\sigma$ and $x \in K$.

Let (E, Π) be a unitary representation of K and $\sigma \in \operatorname{Irr}(K)$. The *multiplicity* of σ in E is $m_\sigma(E) = \dim \operatorname{Hom}_K(V_\sigma, E)$. Here Hom_K consists of the linear maps T with $T\Pi_\sigma(g) = \Pi(g)T$. These maps are bounded because V_σ is finite-dimensional. The *σ-isotypic subspace* of E is the sum of the images of these maps.

Theorem 12.7. *For every $\sigma \in \operatorname{Irr}(K)$,*

$$m_\sigma\big(L^2(K/H)\big) = \dim V_\sigma^H.$$

The σ-isotypic subspace is $E_{\bar\sigma}^H \subseteq C(K/H)$, of dimension $d_\sigma \dim V_\sigma^H$. The space $L^2(K/H)$ is the Hilbert sum of its isotypic subspaces.

Proof. Identify $L^2(K/H)$ with $L^2(K)^H$ through p^*, as in Proposition 12.4. Let $T : V_\sigma \to L^2(K)^H$ be a K-map. By Lemma 12.6 on equivariant images, with $W = \mathbb{C}$, $T(V_\sigma) \subseteq E_{\bar\sigma}$. A vector of $E_{\bar\sigma}$ fixed by A_H lies in $A_H(E_{\bar\sigma}) = E_{\bar\sigma}^H$. Hence

$$\operatorname{Hom}_K\big(V_\sigma, L^2(K/H)\big) = \operatorname{Hom}_K\big(V_\sigma, E_{\bar\sigma}^H\big).$$

Since $V_{\bar\sigma} \cong \overline{V_\sigma}$, we have $\overline{V_{\bar\sigma}} \cong V_\sigma$. So Proposition 12.5 on the decomposition of $L^2(K)^H$ gives

$$E_{\bar\sigma}^H \cong \overline{V_{\bar\sigma}} \otimes V_{\bar\sigma}^H \cong V_\sigma \otimes V_{\bar\sigma}^H,$$

with K acting on the first factor only. By Schur's lemma,

$$\mathrm{Hom}_K\left(V_\sigma, V_\sigma \otimes V_{\bar\sigma}^H\right) \cong V_{\bar\sigma}^H.$$

A K-isomorphism $V_{\bar\sigma} \to \overline{V_\sigma}$ carries $V_{\bar\sigma}^H$ onto $(\overline{V_\sigma})^H$. This is the set V_σ^H with conjugated scalars, so its dimension is $\dim V_\sigma^H$. The images of the K-maps $V_\sigma \to E_{\bar\sigma}^H$ span $E_{\bar\sigma}^H \cong V_\sigma \otimes V_{\bar\sigma}^H$. Finally, $\sigma \mapsto \bar\sigma$ is a bijection of $\mathrm{Irr}(K)$, so the decomposition in Proposition 12.5 gives the Hilbert sum. ∎

Thus the summand E_σ^H of Proposition 12.5 is indexed by σ but is $\bar\sigma$-isotypic. The count is unchanged because $\dim V_\sigma^H = \dim V_{\bar\sigma}^H$. Exercise 12.1 shows this label shift for $K = \mathrm{U}(1)$. Without any bar, the σ-isotypic subspace is spanned by the functions $x \mapsto \langle u, \Pi_\sigma(x)^{-1} v\rangle$ with $u \in V_\sigma^H$ and $v \in V_\sigma$.

12.3 Compact induction and Frobenius reciprocity

Let (τ, W) be a finite-dimensional unitary representation of H. Averaging over H gives every finite-dimensional representation of H an invariant inner product (Lecture 5).

A tangent vector field on S^2 is the model for what follows. Let $K = \mathrm{SO}(3)$, and let H be the subgroup of rotations $R_3(\theta)$ about e_3, as in Lecture 4. Let $x \in \mathrm{SO}(3)$. The columns xe_1 and xe_2 form an oriented orthonormal frame of the tangent plane at $p = xe_3$, and the coset xH is the set of all such frames at p. Let $F(x) \in \mathbb{C}^2$ hold the components of the field at p in this frame. Replacing x by $xR_3(\theta)$ rotates the frame through θ, and the components change by the inverse rotation. So

$$F(xh) = \tau(h)^{-1} F(x), \qquad h \in H,$$

where τ is the rotation action of H on $\mathbb{C}^2$. For example, the tangential part of a fixed vector $v \in \mathbb{R}^3$ has the components $F(x) = (xe_1 \cdot v,\ xe_2 \cdot v)$. Figure 12.2 shows one tangent vector in two frames.

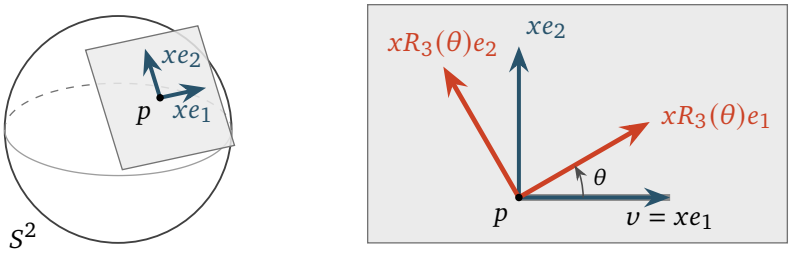


Figure 12.2: On the left, a point $p = xe_3$ of the sphere, with its tangent plane and the frame (xe_1, xe_2). On the right, the same plane seen from outside the sphere, with a second frame for $\theta = \pi/6$, turned anticlockwise. The tangent vector $v = xe_1$ has components $F(x) = (1, 0)$ in the first frame and $F(xR_3(\pi/6)) = (\sqrt{3}/2, -1/2)$ in the second frame.

Definition 12.8. The *induced representation* is

$$\mathrm{Ind}_H^K W = \{F \in C(K, W) : F(xh) = \tau(h)^{-1} F(x) \text{ for all } x \in K,\ h \in H\},$$

with K acting by $(L_g F)(x) = F(g^{-1}x)$.

Left translation preserves the covariance condition because left and right translations commute. For the trivial representation $W = \mathbb{C}$ of H, we recover $C(K)^H \cong C(K/H)$.

On a coset xH the function F is fixed by its value at x. So an element of $\operatorname{Ind}_H^K W$ assigns to each coset a vector of W, once a point of the coset is chosen. In the model above, with $W = \mathbb{C}^2$, the vector $F_1(x)xe_1 + F_2(x)xe_2$ does not depend on the point x of the coset. So $\operatorname{Ind}_H^K W$ is the space of continuous tangent fields on S^2 with complex components. Under this identification, L_g sends a field X to the rotated field $p \mapsto gX(g^{-1}p)$.

For finite K this is the induced representation of the Finite Groups course. The element $x \otimes w$ of $\mathbb{C}[K] \otimes_{\mathbb{C}[H]} W$ corresponds to the function F with $F(xh) = \tau(h)^{-1}w$ for $h \in H$ and $F = 0$ off xH.

Theorem 12.9 (Frobenius reciprocity, continuous form). *Let (Π, V) be a finite-dimensional representation of K. The map*

$$\Phi\colon \operatorname{Hom}_K\left(V, \operatorname{Ind}_H^K W\right) \to \operatorname{Hom}_H(V|_H, W), \qquad \Phi(T)v = (Tv)(e),$$

is a linear bijection. Its inverse sends A to $\Psi(A)$, where $(\Psi(A)v)(x) = A\big(\Pi(x)^{-1}v\big)$.

Proof. First, $\Phi(T)$ is H-equivariant. For $h \in H$, equivariance of T and covariance give

$$\Phi(T)\big(\Pi(h)v\big) = (L_h Tv)(e) = (Tv)(h^{-1}) = \tau(h^{-1})^{-1}(Tv)(e) = \tau(h)\,\Phi(T)v.$$

Next, $\Psi(A)v$ is continuous because $x \mapsto \Pi(x)^{-1}v$ is continuous. Since A is an H-map, $\Psi(A)v$ is covariant:

$$(\Psi(A)v)(xh) = A\big(\Pi(h)^{-1}\Pi(x)^{-1}v\big) = \tau(h)^{-1}A\big(\Pi(x)^{-1}v\big).$$

The map $\Psi(A)$ is a K-map, since $\Pi(x)^{-1}\Pi(g) = \Pi(g^{-1}x)^{-1}$ gives

$$(\Psi(A)\Pi(g)v)(x) = (\Psi(A)v)(g^{-1}x) = (L_g\Psi(A)v)(x).$$

Finally, $\Phi(\Psi(A))v = A(v)$. In the other order,

$$(\Psi(\Phi(T))v)(x) = \big(T\Pi(x^{-1})v\big)(e) = (L_{x^{-1}}Tv)(e) = (Tv)(x). \qquad \blacksquare$$

For finite K, the equality of dimensions is part of Frobenius reciprocity in the Finite Groups course. That course built its bijection on maps out of Ind. Here we build it on maps into Ind, where evaluation at e gives the map directly.

Now pass to L^2. For $h \in H$ define the operator $\widetilde{R}_h = R_h \otimes \tau(h)$ on $L^2(K) \otimes W$. These operators are unitary and commute with $L_g \otimes 1$. For continuous F, $(\widetilde{R}_h F)(x) = \tau(h)F(xh)$. Set

$$L^2\text{-}\operatorname{Ind}_H^K W = \{F \in L^2(K) \otimes W : \widetilde{R}_h F = F \text{ for all } h \in H\}.$$

This closed K-invariant subspace is a unitary representation of K. For the trivial representation $W = \mathbb{C}$, this subspace is $L^2(K)^H$.

Theorem 12.10 (finite multiplicities). *For $\sigma \in \operatorname{Irr}(K)$,*

$$m_\sigma\big(L^2\text{-}\operatorname{Ind}_H^K W\big) = \dim \operatorname{Hom}_H(V_\sigma|_H, W) \le d_\sigma \dim W.$$

Proof. Let $T\colon V_\sigma \to L^2\text{-}\operatorname{Ind}_H^K W$ be a K-map. By Lemma 12.6 on equivariant images, each $F = Tv$ is continuous. Then $\widetilde{R}_h F$ and F are continuous and equal in L^2. Since $C(K)\otimes W \to L^2(K)\otimes W$ is injective, $\tau(h)F(xh) = F(x)$ for all x. So $F \in \operatorname{Ind}_H^K W$. Conversely, continuous covariant functions satisfy $\widetilde{R}_h F = F$. Hence

$$\operatorname{Hom}_K\left(V_\sigma, L^2\text{-}\operatorname{Ind}_H^K W\right) = \operatorname{Hom}_K\left(V_\sigma, \operatorname{Ind}_H^K W\right).$$

Frobenius reciprocity (Theorem 12.9) gives a linear bijection from this space onto $\operatorname{Hom}_H(V_\sigma|_H, W)$, so the dimensions agree. The bound holds because $\operatorname{Hom}_H(V_\sigma|_H, W)$ is a subspace of $\operatorname{Hom}(V_\sigma, W)$, which has dimension $d_\sigma \dim W$. ∎

Remark 12.11 (duals and covariance)**.** For $W = \mathbb{C}$ the count is $\dim \operatorname{Hom}_H(V_\sigma, \mathbb{C}) = \dim(V_\sigma^*)^H$. Since $\langle u, \Pi_\sigma(h)v\rangle = \langle \Pi_\sigma(h)^{-1}u, v\rangle$, the functional $\langle u, \cdot\rangle$ is H-invariant exactly when $u \in V_\sigma^H$. So $u \mapsto \langle u, \cdot\rangle$ is an antilinear bijection $V_\sigma^H \to (V_\sigma^*)^H$. Hence the count agrees with the multiplicity theorem (Theorem 12.7).

The inverse in $\tau(h)^{-1}$ makes the covariance condition compatible with products in H:

$$F(xh_1h_2) = \tau(h_2)^{-1}F(xh_1) = \tau(h_2)^{-1}\tau(h_1)^{-1}F(x) = \tau(h_1h_2)^{-1}F(x).$$

Remark 12.12 (scope)**.** For compact K, we obtained a finite invariant integral. The irreducible representations were finite-dimensional, and Peter–Weyl applied. We proved a count of finite multiplicities, not a general adjunction for Hilbert representations. For noncompact G, the space G/H may carry no invariant measure. Lecture 17 treats that case.

12.4 The sphere $\mathrm{SO}(3)/\mathrm{SO}(2)$

Let $K = \mathrm{SO}(3)$, let e_1, e_2, e_3 be the standard basis of $\mathbb{R}^3$, and let, as in Lecture 4,

$$R_3(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad H = \{R_3(\theta) : \theta \in \mathbb{R}\} \cong \mathrm{SO}(2).$$

The subgroup H is the stabiliser of e_3. Every unit vector is the third column of some rotation, so the orbit map $q(x) = xe_3$ sends K onto S^2. By Lecture 3, q induces a K-equivariant diffeomorphism $K/H \to S^2$. So $F \mapsto F \circ q$ identifies $C(S^2)$ with $C(K)^H$, and $L_g(F\circ q) = (F\circ g^{-1})\circ q$. We define $L^2(S^2)$ through this identification. The invariant integral on S^2 is the usual area integral divided by 4π, because both are rotation-invariant and normalised, and such an integral is unique.

Let E be a finite-dimensional representation of H. An integer k is a *weight* of E if some nonzero $v \in E$ satisfies $R_3(\theta)v = e^{ik\theta}v$ for all $\theta \in \mathbb{R}$. These vectors and 0 form the *weight-k space*. Suppose E has a basis (v_j) with $R_3(\theta)v_j = e^{ik_j\theta}v_j$ for all θ. Then the weight-k space is spanned by the v_j with $k_j = k$. Indeed, a weight-k vector $\sum_j c_j v_j$ satisfies

$$c_j\left(e^{ik_j\theta} - e^{ik\theta}\right) = 0 \qquad \text{for all } \theta,$$

and this forces $c_j = 0$ when $k_j \neq k$.

From Lecture 8, the irreducibles of SO(3) are V_ℓ, $\ell = 0, 1, 2, \ldots$, of dimension $2\ell + 1$ and spin ℓ. In this section an unqualified V_ℓ means the SO(3)-module. The SU(2)-modules of Lecture 8 are always named as such. Choose the covering $\phi = \mathrm{Ad}$ in the ordered basis (X_1, X_2, X_3) of Lecture 4. The explicit calculation there gives

$$\phi\big(\mathrm{diag}(e^{i\theta/2}, e^{-i\theta/2})\big) = R_3(-\theta), \qquad \ker\phi = \{\pm I\}.$$

The pullback $\Pi_\ell \circ \phi$ is the SU(2)-module $V_{2\ell}$. The diagonal element in this display acts on a basis of $V_{2\ell}$ by the scalars $e^{i(\ell-j)\theta}$, $0 \le j \le 2\ell$. Replacing θ by $-\theta$ therefore gives

$$V_\ell \text{ has a basis } (v_k)_{-\ell\le k\le\ell} \text{ with } \Pi_\ell(R_3(\theta))v_k = e^{ik\theta}v_k \text{ for all } \theta \in \mathbb{R}. \tag{12.4}$$

Theorem 12.13. *Let $\mathcal{Y}_\ell \subseteq L^2(S^2)$ be the V_ℓ-isotypic subspace. Then*

$$L^2(S^2) = \widehat{\bigoplus_{\ell\ge 0}} \mathcal{Y}_\ell, \qquad \mathcal{Y}_\ell \cong V_\ell, \qquad \mathcal{Y}_\ell \subseteq C(S^2).$$

Each integral-spin module occurs exactly once.

Proof. The space V_ℓ^H is the weight-zero space. By the weight basis (12.4) it is the line $\mathbb{C}v_0$, so $\dim V_\ell^H = 1$. Apply the multiplicity theorem (Theorem 12.7). ∎

The same weight basis counts tangent fields. In the tangent model, τ has the weights 1 and -1. By Theorem 12.10, V_ℓ occurs twice for $\ell \ge 1$ and not at all for $\ell = 0$. So S^2 carries no nonzero tangent field fixed by all rotations. This statement concerns invariant fields only, and it is not the hairy-ball theorem about a single continuous field.

Let $\mathcal{P}_\ell$ be the space of complex homogeneous polynomials of degree ℓ on $\mathbb{R}^3$, with $(L_g f)(v) = f(g^{-1}v)$. Let $\Delta = \partial_x^2 + \partial_y^2 + \partial_z^2$ and $\mathcal{H}_\ell = \{f \in \mathcal{P}_\ell : \Delta f = 0\}$. The chain rule gives $\Delta(f \circ g) = (\Delta f) \circ g$ for orthogonal g, so $\mathcal{H}_\ell$ is K-invariant. Restriction to S^2 is K-equivariant. Restriction is injective on $\mathcal{P}_\ell$, since $f(rv) = r^\ell f(v)$. Elements of $\mathcal{H}_\ell|_{S^2}$ are called *spherical harmonics* of degree ℓ.

Put $z_\pm = x \pm iy$. Since $R_3(\theta)^{-1} = R_3(-\theta)$,

$$(L_{R_3(\theta)} z_+)(x, y, z) = (\cos\theta\, x + \sin\theta\, y) + i(-\sin\theta\, x + \cos\theta\, y) = e^{-i\theta} z_+.$$

Similarly $L_{R_3(\theta)} z_- = e^{i\theta} z_-$ and $L_{R_3(\theta)} z = z$. The monomials $z_+^a z_-^b z^c$ with $a + b + c = \ell$ form a basis of $\mathcal{P}_\ell$, and for all θ,

$$L_{R_3(\theta)}\, z_+^a z_-^b z^c = e^{i(b-a)\theta}\, z_+^a z_-^b z^c, \qquad |b - a| \le \ell. \tag{12.5}$$

So every weight of $\mathcal{P}_\ell$ satisfies $|k| \le \ell$.

Proposition 12.14. *The space $\mathcal{Y}_\ell$ is spanned by the left translates of $f_\ell = z_+^\ell|_{S^2}$. Hence every element of $\mathcal{Y}_\ell$ is a spherical harmonic of degree ℓ.*

Proof. The polynomial z_+^ℓ is harmonic:

$$\partial_x^2 z_+^\ell + \partial_y^2 z_+^\ell = \ell(\ell-1)z_+^{\ell-2} + i^2\ell(\ell-1)z_+^{\ell-2} = 0.$$

Let U be the span of all $L_g f_\ell$. Then $U \subseteq \mathcal{H}_\ell|_{S^2}$ is finite-dimensional and K-invariant. Decompose U into irreducibles. A copy of V_j in U lies in the V_j-isotypic subspace $\mathcal{Y}_j$. Since $\mathcal{Y}_j \cong V_j$, this copy equals $\mathcal{Y}_j$. Thus $U = \bigoplus_{j \in S} \mathcal{Y}_j$ for a finite set S. Restriction is injective and equivariant, so the weights of U are weights of $\mathcal{P}_\ell$. Since V_j has weight j, this gives $S \subseteq \{0, \dots, \ell\}$.

Write $f_\ell = \sum_{j \in S} y_j$ with $y_j \in \mathcal{Y}_j$. Each $L_{R_3(\theta)}$ preserves each $\mathcal{Y}_j$, and $L_{R_3(\theta)} f_\ell = e^{-i\ell\theta} f_\ell$. Since the sum is direct, comparing components gives $L_{R_3(\theta)} y_j = e^{-i\ell\theta} y_j$ for all θ. For $j < \ell$ the weight $-\ell$ does not occur in V_j, so $y_j = 0$. Hence $f_\ell \in \mathcal{Y}_\ell$ and $U \subseteq \mathcal{Y}_\ell$. Since $U \neq 0$ and $\mathcal{Y}_\ell$ is irreducible, $U = \mathcal{Y}_\ell$. ∎

Figure 12.3 shades the harmonics of the next example by their sign.

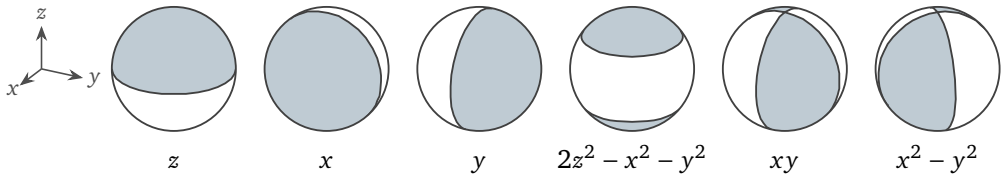


Figure 12.3: Real spherical harmonics of degrees one and two on the unit sphere, shaded where they are positive, with their zero sets drawn. The functions z and $2z^2 - x^2 - y^2$ depend on the latitude only, so they are fixed by H. The second of these vanishes on the latitudes $z = \pm 1/\sqrt{3}$.

Example 12.15 (degrees one and two)**.** We compute $\mathcal{Y}_1$ and $\mathcal{Y}_2$ and their H-fixed lines. Every linear polynomial is harmonic, so $\dim \mathcal{H}_1 = 3$. The map $\Delta\colon \mathcal{P}_2 \to \mathbb{C}$ is onto, since $\Delta(x^2) = 2$. So $\dim \mathcal{H}_2 = 6 - 1 = 5$.

By Proposition 12.14 on spherical harmonics, $\mathcal{Y}_\ell \subseteq \mathcal{H}_\ell|_{S^2}$. Restriction is injective, so $\mathcal{H}_\ell|_{S^2}$ has dimension $2\ell + 1$ for $\ell = 1, 2$. By the sphere theorem (Theorem 12.13), also $\dim \mathcal{Y}_\ell = 2\ell + 1$. So the inclusion is an equality for $\ell = 1, 2$:

$$\mathcal{Y}_1 = \operatorname{span}\{x, y, z\}|_{S^2}, \qquad \mathcal{Y}_2 = \operatorname{span}\{xy,\ yz,\ zx,\ x^2 - y^2,\ 2z^2 - x^2 - y^2\}|_{S^2}.$$

By the monomial weights (12.5), the weight-zero space of $\mathcal{P}_2$ is spanned by $z_+ z_- = x^2 + y^2$ and z^2. Since $\Delta\big(a(x^2 + y^2) + bz^2\big) = 4a + 2b$, its harmonic line is spanned by $2z^2 - x^2 - y^2$. This function spans $\mathcal{Y}_2^H$, the line predicted by $\dim V_2^H = 1$. Similarly z spans $\mathcal{Y}_1^H$. In atomic physics, $\mathcal{Y}_1$ and $\mathcal{Y}_2$ are the angular parts of the p and d orbitals, and on S^2 the H-fixed function $2z^2 - x^2 - y^2 = 3z^2 - 1$ is a multiple of the Legendre polynomial $P_2(z)$.

Exercise 12.5 proves $\mathcal{H}_\ell|_{S^2} = \mathcal{Y}_\ell$ for all ℓ. That exercise defines the Laplacian of the sphere and computes its eigenvalue on $\mathcal{H}_\ell$. Nothing in this lecture depends on it.

12.5 Self-checks

1. Let K be a finite group, regarded as a compact Lie group of dimension zero, and let $H \leq K$ be a subgroup. What does the multiplicity theorem (Theorem 12.7) say?

Answer. Here $L^2(K/H) = \mathbb{C}[K/H]$ is the permutation representation. An irreducible V occurs $\dim V^H$ times. This is finite-group Frobenius reciprocity for the trivial character of H.

2. Why not define $\Phi(T)v = (Tv)(e)$ directly for $T\colon V_\sigma \to L^2\text{-}\operatorname{Ind}_H^K W$?
   Answer. An L^2 class has no value at e. Lemma 12.6 on equivariant images first shows that T takes continuous values.
3. What are the multiplicities for $H = \{e\}$ and for $H = K$?
   Answer. For $H = \{e\}$, σ occurs d_σ times in $L^2(K)$. For $H = K$, a nontrivial irreducible has $V^K = 0$, so only the trivial representation occurs, once, as the constants.

Summary

The invariant integral on K/H is the Haar integral of pullbacks. The averaging operator A_H projects $L^2(K)$ onto $L^2(K)^H \cong L^2(K/H)$. Peter–Weyl then shows that each $\sigma \in \operatorname{Irr}(K)$ occurs in $L^2(K/H)$ exactly $\dim V_\sigma^H$ times.

Let V be a finite-dimensional representation of K and W a finite-dimensional unitary representation of H. An element of $\operatorname{Ind}_H^K W$ is a function on K fixed on each coset by its value at one point, and on S^2 with $W = \mathbb{C}^2$ it is a tangent field with complex components, written in moving frames. Evaluation at e gives $\operatorname{Hom}_K(V, \operatorname{Ind}_H^K W) \cong \operatorname{Hom}_H(V|_H, W)$. Isotypic vectors in $L^2\text{-}\operatorname{Ind}_H^K W$ are continuous. So σ occurs in $L^2\text{-}\operatorname{Ind}_H^K W$ exactly $\dim \operatorname{Hom}_H(V_\sigma|_H, W)$ times.

On $S^2 = \mathrm{SO}(3)/\mathrm{SO}(2)$ each $\mathrm{SO}(3)$-module V_ℓ occurs once in $L^2(S^2)$, realised by spherical harmonics of degree ℓ.

Reading

- Bröcker and tom Dieck, *Representations of Compact Lie Groups* (1985), Section III.6: induction from closed subgroups and Frobenius reciprocity.
- Sepanski, *Compact Lie Groups* (2007), §2.3 and §§3.3–3.4: the assigned background route for this lecture and the sphere example.
- Knapp, *Lie Groups Beyond an Introduction*, digital second edition (2023), Section IX.2: induced representations of compact groups and Frobenius reciprocity.